

Nishant Gadde

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EDUCATION

The University of Texas at Austin

Expected May 2028

B.S. in Electrical and Computer Engineering (Honors) and Mathematics

GPA: 3.91/4.0

- **Coursework:** Probability, Linear Algebra, Numerical Methods, Data Structures & Algorithms, Computer Architecture.
- **4 published/accepted papers** (3 first/co-author) across IEEE ICMLCA, IEEE CAI, and ML/robotics venues.

EXPERIENCE & RESEARCH

Machine Learning Engineer Intern, AI Infrastructure

Jun 2026 – Aug 2026

Apple

Seattle, WA

- Selected as 1 of a small undergraduate cohort for a graduate-scoped role on the AI infrastructure team supporting model serving.
- Working on low-latency inference serving (request batching, caching, accelerator utilization) for production traffic at scale.
- Building tooling for model deployment and serving reliability in a distributed environment handling high queries-per-second.

Research Assistant, Vision-Language Models

Jan 2026 – Present

Oden Institute for Computational Engineering & Sciences, UT Austin

Austin, TX

- Co-developed A4D, a functional-latent-space framework for affordance reasoning in VLMs, reaching **94% inference accuracy** and outperforming the strongest GPT-5-class baseline (79%) by **15+ points** across 8 affordance axes ($\geq 98\%$ per-class).
- Cut new-concept acquisition cost $\sim 10\times$: raised unseen-affordance accuracy from $\sim 70\%$ to **90%** using only **16 labeled examples** ($\sim 10\%$ of training data), robust to 15% label noise.
- Delivered **22 ms** per-inference latency ($\sim 100\times$ faster than prompted VLMs) with isotonic-calibrated uncertainty triggering fallback on $\geq 20\%$ of cases at 93% accuracy; co-authored paper submitted to CoRL 2026.

Machine Learning Engineer Intern

Jun 2025 – Aug 2025

Rayfield Systems

Houston, TX

- Built LSTM time-series forecasting models in PyTorch for high-frequency sensor signals, cutting error $\sim 30\%$ vs. regression baselines.
- Deployed streaming inference on Spark and Kafka and applied weight quantization to reduce inference memory $\sim 4\times$ on edge devices.

Research Assistant

Jun 2024 – Aug 2025

Stanford Sustainability Systems Laboratory (S3L)

Remote

- Co-authored an IEEE CAI paper approximating non-convex AC optimal power flow with deep networks, achieving **1% optimality gap** at $\sim 1000\times$ lower solve time than a conventional solver.
- Enforced feasibility via SOCP convex relaxations during training and applied PCA to compress grid features, improving generalization.

Research Intern

Jun 2024 – Dec 2024

Cardiovascular Fluid Mechanics Laboratory, Georgia Tech

Atlanta, GA

- Built a hybrid CNN+SVM ECG pipeline on PhysioNet signals, hitting **100% accuracy** on both training and validation sets under stratified K-fold cross-validation.
- Engineered temporal features (QRS duration, RR intervals) and FFT frequency-domain features; co-authored paper accepted at **IEEE ICMLCA 2025**.

PROJECTS

C++ Low-Latency Order Book Engine | C++17, lock-free queues, Linux perf, Valgrind

2026

- Implemented a price-level limit order book with **O(1)** insert/cancel using lock-free ring buffers, sustaining **1M order messages/sec** in replay.
- Modeled market microstructure under Poisson order arrivals: adverse selection cost, queue-priority decay, and passive fill probability.
- Profiled with **perf** and Valgrind to eliminate false sharing and hot-path heap allocation, cutting **p99.9 latency** $\sim 40\%$ to **sub-microsecond**.
- Built a market-making layer quoting two-sided prices with inventory-risk controls and adverse-fill-aware quote skewing.

Statistical Arbitrage Engine | Python, C++, NumPy, statsmodels, SciPy

2026

- Screened **500+ equity pairs** for cointegration via Johansen tests and fit spreads to an Ornstein-Uhlenbeck process by MLE for entry signals.
- Estimated dynamic hedge ratios with a Kalman filter using adaptive process-noise covariance, reducing beta drift $\sim 35\%$ across regimes.
- Validated with walk-forward out-of-sample testing under modeled costs (bid-ask spread, slippage, short-borrow), reaching **out-of-sample Sharpe ≈ 1.8** .
- Tracked per-trade PnL attribution and risk (max drawdown, rolling Sharpe, exposure) to separate alpha from execution and beta effects.

Monte Carlo Derivatives Pricer | C++17, NumPy, variance reduction

2025

- Priced European and path-dependent options via **1M+ path** Monte Carlo simulation of geometric Brownian motion.
- Cut estimator variance $\sim 10\times$ with antithetic and control variates; computed Greeks via pathwise and finite-difference methods.
- Benchmarked against closed-form Black-Scholes, converging to **0.5% pricing error** to validate convergence and quantify discretization error.

HONORS & COMPETITIONS

- USACO Platinum Division, USA Computing Olympiad2024–2025
- Qualified for the Platinum division (USACO’s **top tier**, ~**top 1%**), solving advanced graph, DP, and data-structure problems.
 - Ranked **top 100 worldwide** in the 2024–25 competitive-programming season (December - US Open).

TECHNICAL SKILLS

Programming: C++ (*primary*), Python, SQL, Java, Rust, MATLAB, Bash
Math & Quant: Probability & Statistics, Linear Algebra, Stochastic Calculus, Optimization, Numerical Methods
ML & Systems: PyTorch, TensorFlow, XGBoost, Spark, Kafka, Docker, Linux, Git, CMake
Quant Methods: Market Microstructure, Time-Series Analysis, Backtesting, Monte Carlo, Kalman Filtering, Cointegration
Finance: Options & Derivatives Pricing, Greeks, PnL Attribution, Risk Metrics (VaR, Sharpe), Mean Reversion
Libraries: NumPy, pandas, SciPy, statsmodels, scikit-learn